

Yanbo Zhou

Ph.D. Candidate
Non-Volatile Systems Lab (NVSL)
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Research Interests

My research focuses on computer systems for cloud and AI infrastructure, with a particular emphasis on memory and storage systems. My goal is to make computer systems faster, more resource-efficient, and more reliable for traditional and agentic cloud workloads at large scale.

Education

09/2023 - Present	University of California, San Diego Ph.D. in Computer Science Advisor: Prof. Steven Swanson	San Diego, USA
09/2016 - 01/2019	East China Normal University M.Eng. in Software Engineering	Shanghai, China
09/2012 - 06/2016	Taiyuan University of Technology B.Eng. in Software Engineering	Taiyuan, China

Publications

Refereed Conference and Journal Publications

TOS'26	Cloud Local Storage Goes Elastic: From Hardware Offloading to Hybrid Architecture Leping Yang, Yanbo Zhou , Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue 2026 ACM Transactions on Storage (TOS)
OSDI'26	Break on Through to the Other Side: Pooling Memory Elastically with RamRyder Yanbo Zhou , Erci Xu, Dongjoo Seo, Adam Manzanares, Steven Swanson. 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI) [paper] [project]
FAST'26	Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud Leping Yang, Yanbo Zhou , Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue 24th USENIX Conference on File and Storage Technologies (FAST) [paper] Erik Riedel Best Paper Award
SOSP'25	Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman Yanbo Zhou , Erci Xu, Anisa Su, Jim Harris, Adam Manzanares, Steven Swanson. 31st ACM Symposium on Operating Systems Principles (SOSP) [paper]
EuroSys'24	CSAL: the Next-Gen Local Disks for the Cloud Yanbo Zhou , Erci Xu, Li Zhang, Kapil Karkra, Mariusz Barczak, Wayne Gao, Wojciech Malikowski, Mateusz Kozłowski, Łukasz Łasek, Ruiming Lu, Feng Yang, Lilong Huang, Xiaolu Zhang, Wenrui Li, Jinhu Li, Keqiang Niu, Jiaji Zhu, Jiesheng Wu. 19th ACM European Conference on Computer Systems (EuroSys) [paper] [project]

- ATC'20 **Spool: Reliable Virtualized NVMe Storage Pool in Public Cloud Infrastructure.**
Shuai Xue, Shang Zhao, Quan Chen, Gang Deng, Zheng Liu, Jie Zhang, Zhuo Song, Tao Ma, Yong Yang, **Yanbo Zhou**, Keqiang Niu, Sijie Sun, Minyi Guo.
2020 USENIX Annual Technical Conference (ATC) [[paper](#)]
- RTCSA'18 **Write-aware Data Allocation on Heterogeneous Memory Architecture with Minimum Cost.**
Yanbo Zhou, Shouzheng Gu, Lixia Zheng, Edwin H.-M. Sha, Qingfeng Zhuge, Lin Wu.
24th IEEE International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA) [[paper](#)]
- SC2'18 **Accelerating I/Os in Virtual Machines on Physical NVMe SSDs via User Space Vhost Target.**
Ziye Yang, Changpeng Liu, **Yanbo Zhou**, Xiaodong Liu, Gang Cao
8th IEEE International Symposium on Cloud and Service Computing (SC2) [[paper](#)] [[project](#)]

Preprints

- 2026 **SmoothAgent: Efficient Long-Horizon LLM-Based Agent Serving with Lookahead Context Engineering**
Zaifeng Pan, Qianxu Wang, Zhengding Hu, Chang Chen, Yue Guan, **Yanbo Zhou**, Steven Swanson, and Yufei Ding.
arXiv preprint arXiv:2607.00151 [[paper](#)]

Tech Reports

- 2023 **A Media-Aware Cloud Storage Acceleration Layer (CSAL) Cache Solution with Intel Optane SSDs for Alibaba ECS Local Disk D3C Service.**
Yanbo Zhou, Li Zhang, Kapil Karkra, Wayne Gao, Chunhong Mao, Mariusz Barczak.
Intel White Paper [[paper](#)].

Book Chapters

- 2019 **Linux Open Source Storage: from Ceph to Container (Chinese Edition)**
Published by the Publishing House of Electronics Industry [[book](#)]

Talks

- Break on Through to the Other Side: Pooling Memory Elastically with RamRyder**
07/2026 OSDI'26
06/2026 UCLA System Seminar
05/2026 SRC ACE Center for Evolvable Computing [[slides](#)] [[talk](#)]
- Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud**
04/2026 Meta AI and Systems Co-Design Team
02/2026 FAST'26 [[slides](#)] [[talk](#)]
- Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman**
12/2025 SRC ACE Center for Evolvable Computing
09/2025 Meta AI and Systems Co-Design Team [[slides](#)]
- CSAL: the Next-Gen Local Disks for the Cloud**
04/2025 EuroSys'24 [[slides](#)]
- Best SPDK Practices: Lessons from Five Years of Storage Evolution in Alibaba Cloud**

12/2022 SPDK PRC Forum [[slides](#)] [[talk - Chinese](#)]

Removing QLC Write-Amplification through Intel Optane SSD with SPDK WSR

12/2021 SPDK PRC Forum [[slides](#)] [[talk - Chinese](#)]

Academic Service

Reviewer

2026 IEEE Computer Architecture Letters (CAL)

Artifact Evaluation Committee

2026 USENIX FAST, USENIX OSDI

2025 ACM ASPLOS, ACM EuroSys, ACM SOSP

Professional Experience

09/2023 - **University of California, San Diego** San Diego, USA
Present *Graduate Student Researcher, Non-Volatile Systems Lab (NVSL)*

Led research on memory and storage systems for future cloud and AI agent infrastructure. The research work has been published at SOSP'25 and OSDI'26.

Summers **Samsung Semiconductor** San Jose, USA
 24/25/26 *Software Engineering Intern, Global Open-ecoSystem Team (GOST)*

Worked on profiling emerging memory and storage hardware, such as CXL memory and large-capacity SSDs, and identified system-level bottlenecks.

01/2019 - 06/2023	Alibaba Cloud <i>Senior Software Engineer, Elastic Block Storage (EBS) Team</i>	Hangzhou, China
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Led research and development of block storage systems in the cloud to address challenges of deployed systems at scale and advance cloud storage systems. The research work has been published at USENIX ATC'20, EuroSys'24, and FAST'26.

Jul 2017 - **Intel Corporation** Shanghai, China
Dec 2018 *Software Engineering Intern, Storage Performance Development Kit (SPDK) Team*

Worked on SPDK project and contributed to the polling-based kernel-bypass NVMe virtualization solution Vhost-NVMe and tools for kernel-bypass storage stack.

Teaching Experience

Fall 2025 **Introduction to Computer Architecture: A Software Perspective (UCSD CSE 142)**
Teaching Assistant

Head TA for an undergraduate computer architecture course with 90+ students; led four tutors and collaborated with instructor (Prof. Steven Swanson) to design assignments, quizzes, and exams. Held office hours to help students debug and reason through technical issues related to computer architecture.

Mentoring

2025-2026 **Leping Yang**, Ph.D. student at Shanghai Jiao Tong University

Achievements: First-author papers at FAST'26 (Best Paper Award) and TOS'26.

2019-2020 **Shuai Xue**, M.S. student at Shanghai Jiao Tong University (now at Alibaba Cloud)

Achievements: First-author paper at USENIX ATC'20.